

Durwas Khot

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SUMMARY

I am an Automotive engineering postgraduate with hands-on Vehicle dynamics experience gained through Oxford Brookes projects and a Bentley Motors Master's thesis. Experienced in 3D CAD, Rigid/Flexible Multi-Body Dynamics simulations and cross-functional vehicle development. I am passionate about solving complex engineering problems and have a drive to deliver exceptional ride performance in next-generation vehicles.

EDUCATION

MSc Automotive Engineering with EVs

Oxford Brookes University

Distinction (1:1)

September 2024 - September 2025

BE - Mechanical Engineering

University of Mumbai

77.60% (UK Equivalent – 1:1)

August 2019 - May 2023

RESEARCH AND INDUSTRY PROJECTS

Oxford Brookes University (In collaboration with Bentley Motors, UK)

Master's Thesis Student

May 2025 – September 2025

- Developed flexible MBD models in MSc Adams and PINN-based characterisation for steering system bushes in limited availability of test data, achieving a 25% improvement in bushing stiffness/damping correlation.
- Performed NVH data analysis across different test cases, delivering design insights that reduced steering VTV from 2.38 to 1.64 m/s² through non-linear bushing integration, validated against ISO 5349 comfort metrics.
- Conducted correlation of high-frequency (above 80 Hz) road-induced vibration in electric vehicle systems, through test and flexible multi-body dynamics simulation to improve model prediction accuracy by 30%.

Oxford Brookes University

September 2024 – December 2024

- Developed and validated twin-track bicycle models in MSC Adams and analysed lateral dynamics, yaw rate response, slip angle behaviour, and lateral load transfer for a 2025 Toyota Camry in different driving conditions.
- Modelled a 4-DOF suspension with damper dyno test data to study ride comfort, body displacement, pitch motion, and frequency response. Identified a damping ratio of 0.2 and spring stiffness parameters to minimise contact patch variation and validated resonant frequency analytically (1.38 Hz) against simulation results (1.3 Hz).

Formula Student team - Modified Auto Club Racing, Thane (Founding member)

Team Manager and Suspension Lead

January 2022 – December 2022

Head - Vehicle Dynamics Department

May 2021 - December 2021

- Delivered a Best Debut Team Award by leading a 30-member engineering team through the design, prototyping, manufacturing, and system-level validation of an FS car, coordinating subsystem integration and compliance.
- Improved cornering stability by optimising suspension and chassis kinematics including ± 40 mm wheel travel, 1.5° camber gain during bump, and 35-45mm roll height through Multi-body Dynamics (MBD) simulation and CAD modelling in CATIA V5, taking ownership of design from concept through build.

WORK EXPERIENCE

Junior Design Engineer

June 2023 – August 2024

TRITORC Equipments Pvt. Ltd.

- Designed and developed industrial mechanical systems, including a 72-inch diameter cold cutting machine, ensuring structural integrity and validation of operational reliability. Generated 3D CAD models, 2D drawings, and FEA with the implementation of DFMEA and PFMEA.
- Led collaboration with cross-functional teams such as manufacturing, procurement, and testing to optimise cost/performance ratios, design upgrades, and reduce manufacturing complexity by iterative design reviews.

SKILLS

Software Tools: CATIA, SolidWorks, Ansys, MATLAB, Simulink/Simscape, MSC Adams, Python programming

Engineering: CAD/CAE, GD&T, FEA, Rigid/Flexible Multi-Body Dynamics, Technical Documentation, Vehicle Dynamics modelling, Mathematics, Engineering Physics

Management: Engineering Project Management, IATF 16949, ISO 9001 QMS, ISO 14001 EMS, Lean Principles, 8D Methodology

Soft Skills: Analytical thinking, Leadership, Collaboration, Quality-driven, Problem-solving, Self-motivation